

ASSIGNMENT -2

CO3: EXCEPTION HANDLING AND MULTITHREADING

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COURSE CODE: CSA0920

COURSE NAME: PROGRAMMING IN JAVA

ASSIGNMENT SET: DIRECT-DESCRIPTIVE

Q1. Basic Try-Catch Block

Write a Java program that accepts two integers from the user and divides the first by the second. Wrap the division operation in a try block, and catch the Arithmetic Exception that occurs when the divisor is zero, displaying a user-friendly error message instead of letting the program crash. Test the program with both valid and zero-divisor inputs.

Concepts: try, catch, Arithmetic Exception.

GitHub Repository Link: <https://github.com/prabhath191006/Java>

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2. PROBLEM UNDERSTANDING

The program accepts two integers from the user and performs division using the first number as the dividend and the second as the divisor. It must handle the division-by-zero situation without crashing by using Java's exception-handling mechanism.

The main concepts demonstrated are **try, catch, and Arithmetic Exception**.

3. APPROACH / DESIGN NOTES

- Used a single Assignment2 class containing the main () method.
- Used the Scanner class to accept two integer inputs from the user.
- Used catch (Arithmetic Exception e) to handle division by zero.

4. SOURCE CODE

```
import java.util.Scanner;

public class Assignment2 {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter the first integer: ");

        int firstNumber = scanner.nextInt();

        System.out.print("Enter the second integer: ");

        int secondNumber = scanner.nextInt();

        try {

            int result = firstNumber / secondNumber;

            System.out.println("Result: " + result);

        } catch (ArithmeticException e) {

            System.out.println("Error: Cannot divide by zero.");
```

```

    }

    scanner.close();

}

}

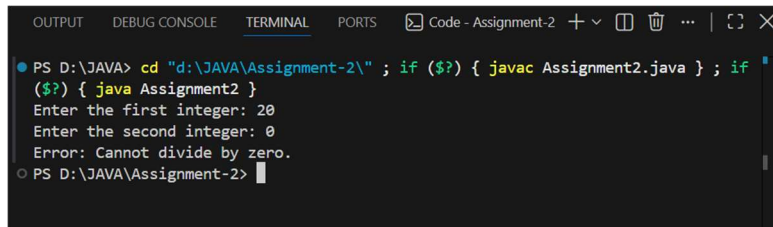
```

5. TEST CASES

S.NO	Input	Expected Output	Actual Output
1 – Normal case	First = 20, Second = 5	Result: 4	Result: 4
2 – Edge case	First = 20, Second = 0	Error: Cannot divide by zero.	Error: Cannot divide by zero.
3 – Boundary case	First = 0, Second = 5	Result: 0	Result: 0

6. SCREENSHOTS OF EXECUTION

TEST CASE - 1

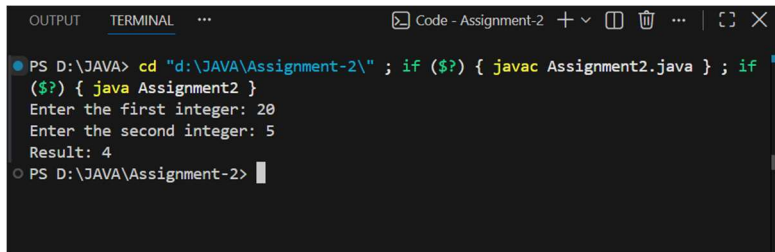


```

PS D:\JAVA> cd "d:\JAVA\Assignment-2\" ; if ($?) { javac Assignment2.java } ; if ($?) { java Assignment2 }
Enter the first integer: 20
Enter the second integer: 0
Error: Cannot divide by zero.
PS D:\JAVA\Assignment-2>

```

TEST CASE - 2



```
PS D:\JAVA> cd "d:\JAVA\Assignment-2\" ; if ($?) { javac Assignment2.java } ; if ($?) { java Assignment2 }
Enter the first integer: 20
Enter the second integer: 5
Result: 4
PS D:\JAVA\Assignment-2>
```

TEST CASE - 3



```
PS D:\JAVA> cd "d:\JAVA\Assignment-2\" ; if ($?) { javac Assignment2.java } ; if ($?) { java Assignment2 }
Enter the first integer: 0
Enter the second integer: 5
Result: 0
PS D:\JAVA\Assignment-2>
```

7. REFLECTION

I learned how Java exception handling can prevent a program from terminating unexpectedly when an arithmetic error occurs. I also understood how the try-catch mechanism can handle Arithmetic Exception and provide a user-friendly message for division by zero.

8. DECLARATION

This is my own work. Any AI/external tool assistance used is disclosed above.